

Advanced BERT Spam Detection

```
!pip install transformers datasets evaluate
```

```
import torch
import numpy as np
import pandas as pd
from datasets import load_dataset
from transformers import (
    BertTokenizer,
    BertForSequenceClassification,
    TrainingArguments,
    Trainer
)
from sklearn.metrics import accuracy_score, precision_recall_fscore_support
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
Collecting evaluate
  Downloading evaluate-0.4.6-py3-none-any.whl.metadata (9.5 kB)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.21.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.3.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
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Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.0)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.23.1)
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Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
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Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.3.0) (2025.3.0)
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Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (3.4.0)
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Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0) (0.14.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->dataset) (1.17.0)
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Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer>=0.23.1->typer-slim) (0.0.2)
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Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=10.11.0) (0.1.2)
Downloading evaluate-0.4.6-py3-none-any.whl (84 kB)
84.1/84.1 kB 1.9 MB/s eta 0:00:00
Installing collected packages: evaluate
Successfully installed evaluate-0.4.6
```

```
df = pd.read_csv("spam_r.csv")
```

```
df['label'] = df['label'].map({'ham': 0, 'spam': 1})
```

```
tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')
```

```
def tokenize_function(examples):
    return tokenizer(
        examples['text'],
```

```
padding='max_length',
truncation=True,
max_length=128
)
```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF\_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.

```
warnings.warn(
tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 3.81kB/s]
vocab.txt: 100% 232k/232k [00:00<00:00, 2.88MB/s]
tokenizer.json: 100% 466k/466k [00:00<00:00, 11.8MB/s]
```

```
from datasets import Dataset

dataset = Dataset.from_pandas(df)
dataset = dataset.train_test_split(test_size=0.2)

dataset = dataset.map(tokenize_function, batched=True)
dataset = dataset.rename_column("label", "labels")
dataset.set_format('torch', columns=['input_ids', 'attention_mask', 'labels'])
```

```
Map: 100% 16/16 [00:00<00:00, 153.28 examples/s]
Map: 100% 4/4 [00:00<00:00, 44.79 examples/s]
```

```
model = BertForSequenceClassification.from_pretrained('bert-base-uncased', num_labels=2)
```

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
config.json: 100% 570/570 [00:00<00:00, 16.6kB/s]

```
model.safetensors: 100% 440M/440M [00:06<00:00, 91.5MB/s]
Loading weights: 100% 199/199 [00:00<00:00, 237.93it/s, Materializing param=bert.pooler.dense.weight]
```

BertForSequenceClassification LOAD REPORT from: bert-base-uncased

Key	Status
cls.predictions.transform.dense.bias	UNEXPECTED
cls.seq_relationship.weight	UNEXPECTED
cls.predictions.transform.LayerNorm.bias	UNEXPECTED
cls.predictions.bias	UNEXPECTED
cls.predictions.transform.dense.weight	UNEXPECTED
cls.predictions.transform.LayerNorm.weight	UNEXPECTED
cls.seq_relationship.bias	UNEXPECTED
classifier.weight	MISSING
classifier.bias	MISSING

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING :those params were newly initialized because missing from the checkpoint. Consider training on your downstream t

```
def compute_metrics(pred):
    labels = pred.label_ids
    preds = pred.predictions.argmax(-1)

    precision, recall, f1, _ = precision_recall_fscore_support(labels, preds, average='binary')

    acc = accuracy_score(labels, preds)

    return {
        "accuracy": acc,
        "precision": precision,
        "recall": recall,
        "f1": f1,
    }
```

```
import transformers
from transformers import TrainingArguments

print("Transformers version:", transformers.__version__)
print("TrainingArguments location:", TrainingArguments)
```

Transformers version: 5.0.0
 TrainingArguments location: <class 'transformers.training_args.TrainingArguments'>

```
training_args = TrainingArguments(
    output_dir="./results",
    eval_strategy="epoch",
    save_strategy="epoch",
    learning_rate=2e-5,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    num_train_epochs=4,
    weight_decay=0.01,
    warmup_ratio=0.1,
    logging_dir="./logs",
    load_best_model_at_end=True,
    metric_for_best_model="f1",
    gradient_accumulation_steps=2,
    fp16=True
)
```

warmup_ratio is deprecated and will be removed in v5.2. Use `warmup\_steps` instead.
 `logging\_dir` is deprecated and will be removed in v5.2. Please set `TENSORBOARD\_LOGGING\_DIR` instead.

```
from transformers import TrainingArguments
print(TrainingArguments)
```

<class 'transformers.training_args.TrainingArguments'>

```
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=dataset["train"],
    eval_dataset=dataset["test"],
    compute_metrics=compute_metrics
)
```

```
trainer.train()
```

/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:668: UserWarning: 'pin_memory' argument is set as true but no pin_memory_device is provided. This warning can be disabled by setting 'warn_pin_memory' to False.
 warnings.warn(warn_msg)

[4/4 04:26, Epoch 4/4]

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	No log	0.757556	0.500000	0.500000	1.000000	0.666667
2	No log	0.734979	0.500000	0.500000	1.000000	0.666667
3	No log	0.718602	0.500000	0.500000	1.000000	0.666667
4	No log	0.709143	0.500000	0.500000	1.000000	0.666667

Writing model shards: 100% 1/1 [00:05<00:00, 5.27s/it]

/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:668: UserWarning: 'pin_memory' argument is set as true but no pin_memory_device is provided. This warning can be disabled by setting 'warn_pin_memory' to False.
 warnings.warn(warn_msg)

Writing model shards: 100% 1/1 [00:05<00:00, 5.90s/it]

/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:668: UserWarning: 'pin_memory' argument is set as true but no pin_memory_device is provided. This warning can be disabled by setting 'warn_pin_memory' to False.
 warnings.warn(warn_msg)

Writing model shards: 100% 1/1 [00:21<00:00, 21.48s/it]

/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:668: UserWarning: 'pin_memory' argument is set as true but no pin_memory_device is provided. This warning can be disabled by setting 'warn_pin_memory' to False.
 warnings.warn(warn_msg)

Writing model shards: 100% 1/1 [00:22<00:00, 22.82s/it]

There were missing keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.weight', 'bert.embeddings.LayerNorm.bias']
 There were unexpected keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.beta', 'bert.embeddings.LayerNorm.gammas']
 TrainOutput(global_step=4, training_loss=0.6863815188407898, metrics={'train_runtime': 305.5147, 'train_samples_per_second': 0.209, 'train_steps_per_second': 0.013, 'total_flos': 4209776885760.0, 'train_loss': 0.6863815188407898, 'epoch': 4.0})

